

Synthesis of a Novel 1,3,4-Oxadiazole Derivative with Piperazine and Phenoxypropanol Functionalities

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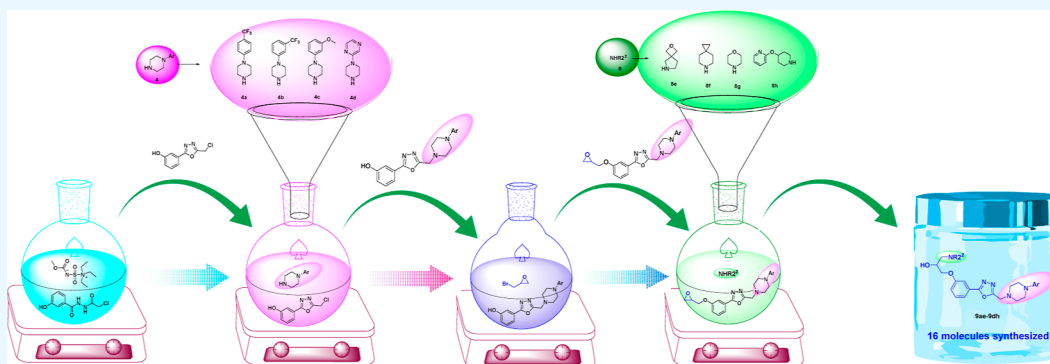
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ABSTRACT: Synthesis of novel heterocyclic derivatives of 1-amino-3-(3-(5-((4-phenylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol is focused on specific piperazine and piperidine derivatives of 1,3,4-oxadiazole. These compounds are a distinguished motif for biological studies. An effective piperazine and piperidine derivatives methodology on 1,3,4-oxadiazole substrates has been developed. This pathway along with good yields and clean reactions has a broad substrate scope. This pathway provides new biologically useful moieties that can further be investigated for their applicability.

1. INTRODUCTION

Oxadiazoles are of considerable interest in diverse areas of medicinal chemistry, pesticides, polymer, and material science¹ Oxadiazole-containing compounds have had a large impact on multiple drug discovery programs across a variety of disease areas including diabetes,² obesity³ inflammation⁴ cancer⁵ and infection⁶ Oxadiazole rings are utilized in drug discovery programs for several different purposes. In some cases, they are an essential part of the pharmacophore contributing favorably to ligand binding⁷ In other cases, oxadiazole moieties have been shown to act as flat, aromatic linkers to place substituents in the appropriate orientation⁸ as well as to modulate molecular properties by positioning them in the periphery of the molecule⁹ Oxadiazoles display interesting hydrogen bond acceptor properties and it has been shown that different Regioisomers exhibit significantly different hydrogen bonding behaviors¹⁰ They have also been employed as replacements for carbonyl and carboxy groups, such as amides, carbamates, and hydroxamic esters^{11–13} Among the various isomeric oxadiazoles, the 1,2,4-oxadiazole, an ester isostere, is present in various biologically active compounds, such as benzodiazepine receptor ligands, muscarinic receptor agonists, and 5-HT₃ receptor antagonists.⁹ Aryl and heteroaryl-substituted 1,2,4-oxadiazole-5-carboxamides were recently reported as antiplatelet and antithrombotic agents, partial serotonin antagonists,¹⁴ and have also been utilized as β -amyloid imaging agents

for Alzheimer's disease⁸ Despite their widespread use, the chemistry to prepare these compounds remains challenging specifically, the small aza rich intermediates can present a significant thermal hazard.

Some examples which illustrate the diverse biological activities of heterocyclic nucleus 1,3,4-oxadiazole are Furamizole, 1,2-amino-5-[2-(5-nitro-2-furyl)-1-(2-furyl)-vinyl]1,3,4-oxadiazole, a nitrofuran derivative possesses a strong antibacterial activity^{15–18}

Nesapidil, 2,1-[4-(2-methoxyphenyl)piperazin-1-yl]-3-[3-(5-methyl-1,3,4-oxadiazol-2-yl)phenoxy]propan-2-ol, a well-established vasodilating agent also contains 1,3,4-oxadiazole nucleus. It is calcium channel blocker. Its major effect is to slow down Ca²⁺ channels¹⁹

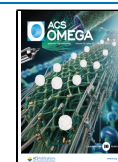
Other molecule (3A) also synthesized by Ibrahim et al. for cardiovascular diseases such as atrial and ventricular arrhythmias, intermittent claudication, Prinzmetal's variant angina, stable and unstable angina.²⁰ Figure 1.

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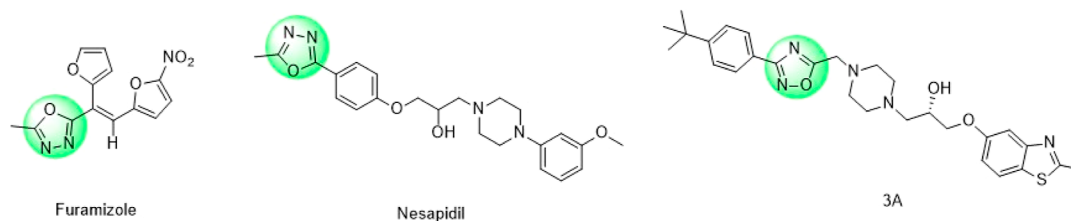
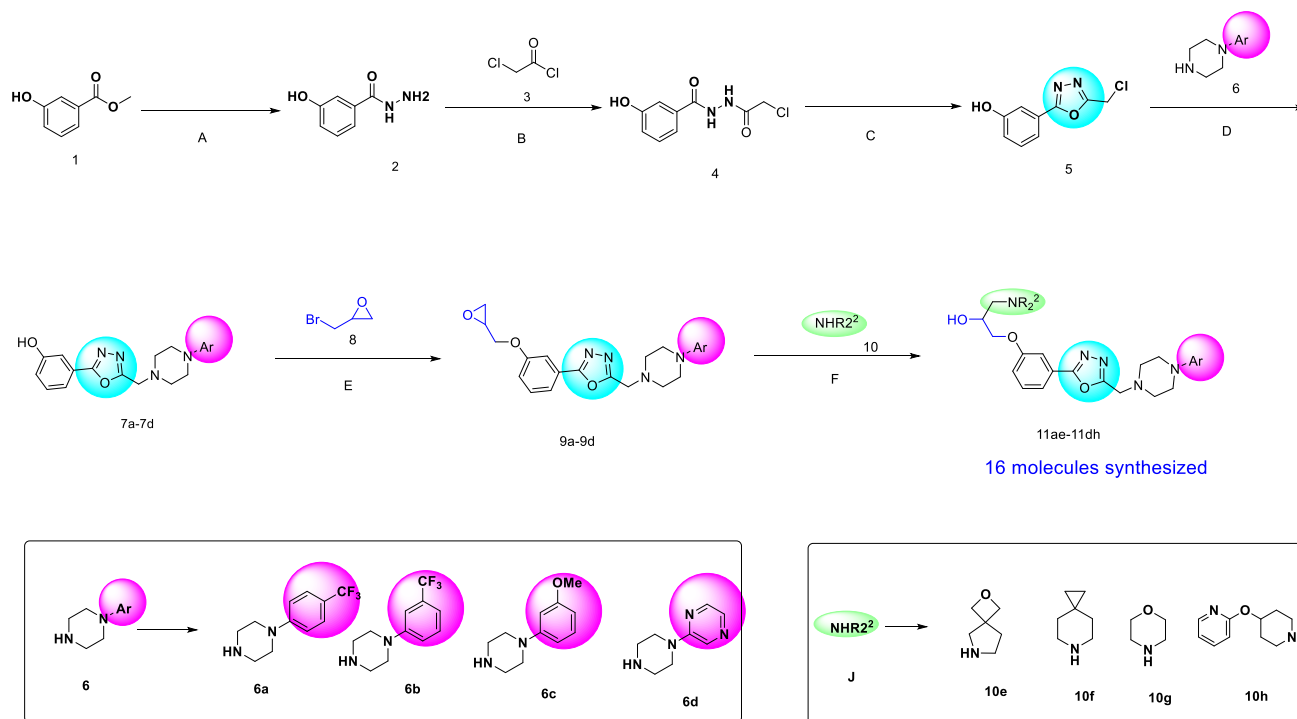


Figure 1. Biological active of heterocyclic nucleus.

Scheme 1. General Synthetic Scheme for the Synthesis of 1-Amino-3-(3-(5-(piperazin-1-ylmethyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol^a



^a(A) $\text{NH}_2\text{NH}_2 \cdot \text{H}_2\text{O}$ (5 equiv), IPA, 90 °C, 16 h, 97%; (B) Chloroacetyl chloride (1.1 equiv), THF, 0 °C–rt, 16 h 95%; (C) Burgess reagent (1.5 equiv), dioxane, 100 °C, 16 h, 76%. (D) Pyridine (3 equiv), THF, rt, 16 h, 81%. (E) NaH (2 equiv), THF, rt, 2h, 87%; (E) K_2CO_3 , ACN, 90 °C, 3–8 h, 65–94%.

In the present invention relates to novel heterocyclic derivatives, in particular piperazine and piperidine derivatives of oxadiazole, we focused on the synthesis of these and its covers a wide range of substrate scope.

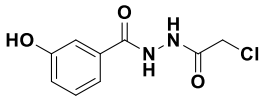
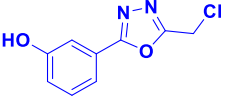
2. RESULT AND DISCUSSION

We report a general and novel synthesis of 3-(3-(5-(piperazin-1-ylmethyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol amines in good yields. Our study began with the synthesis of 3-hydroxybenzohydrazide (2), which was converted to *N'*-(2-chloroacetyl)-3-hydroxybenzohydrazide (4). This intermediate was then cyclized using Burgess reagent to form 3-(5-(chloromethyl)-1,3,4-oxadiazol-2-yl)phenol (5). The resulting compound was treated with piperazine aryl derivatives to obtain the corresponding 3-(5-(4-aryl-1-piperazinylmethyl)-1,3,4-oxadiazol-3-yl)phenol derivatives (7a–7d) in good yields. These compounds were then treated with epibromohydrin, leading to a Williamson ether synthesis of the corresponding alkyl aryl ethers (9a–9d). Finally, the epoxy ring was opened by reaction with secondary amines, yielding the target 3-(3-(5-(piperazin-1-ylmethyl)-1,3,4-oxadiazol-3-yl)-

phenoxy)propan-2-ol amines in good yields. The reaction strategy is depicted in Scheme 1.

Methyl 3-hydroxybenzoate was treated with hydrazine hydrate (5 equiv) in IPA at 90 °C for 12 h to obtain 3-hydroxybenzohydrazide (2). This compound was then treated with chloroacetyl chloride in THF at 0 °C to room temperature for 16 h, resulting in the formation of *N'*-(2-chloroacetyl)-3-hydroxybenzohydrazide (4). After successfully synthesizing compound (4), it was subjected to different optimization conditions to identify the most effective reaction method for the synthesis of 3-(5-(chloromethyl)-1,3,4-oxadiazol-2-yl)phenol (5) (Table 1). The cyclodehydration of 1,2-diacylhydrazines typically requires harsh reagents, such as e.g. SOCl_2 ,²¹ POCl_3 ,²² polyphosphoric acid,²³ or sulfuric acid.²⁴ POCl_3 in ACN at 100 °C provides a 61% yield of cyclodehydration. These reactions were conducted under an inert atmosphere of argon in oven-dried round-bottom flasks (RBF).²⁵ We speculated that Burgess reagent may provide a suitable alternative. Burgess reagent is the reagent of choice for the cyclodehydration of hydroxamides and thioamides to azoles.²⁶ Additionally, dehydration of primary nitroalkanes to

Table 1. Optimization of the Reaction Conditions for the Synthesis of 3-(5-(Chloromethyl)-1,3,4-oxadiazol-3-yl)phenol^a

 4 Dehydrating agent (1.5–5 equiv), solvent, 70–140 °C, 4–36h C  5							
entry	dehydrating agent	equiv	solvent	temp, °C	time, h	Int-4 (%)	Int-5 (%) yield
1	POCl ₃	2	ACN	70	24		31
2	POCl ₃		ACN	100	24		61
3	POCl ₃			110	16		9
4	SOCl ₂			110	16		8
5	Burgess reagent	2	THF	rt	16	11	traces
6	Burgess reagent	5	THF	100	16	14	12
7	Burgess reagent	2	THF	140	16	8	traces
8	Burgess reagent	1.5	dioxane	100	4	9	12
9	Burgess reagent	1.5	dioxane	100	24		76
10	Burgess reagent	1.5	dioxane	140	16	6	14
11	Burgess reagent	2	dioxane	100	36	12	65
12	Burgess reagent	5	dioxane	100	16	9	4
13	Burgess reagent	1.5	DMSO	100	4		traces
14	Burgess reagent	1.5	ACN	100	16	38	2
15	Burgess reagent	1.5	toluene	100	16	80	3

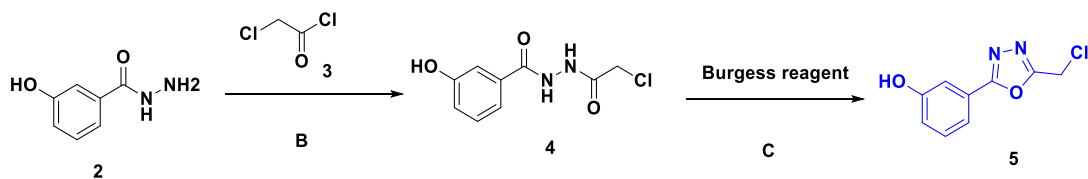
^aReaction conditions: dehydrating agent (1.5–5 equiv), solvent, 70–140 °C, solvent, 4–36 h.

Figure 2. Synthesis of 3-(5-(chloromethyl)-1,3,4-oxadiazol-3-yl)phenol (5).

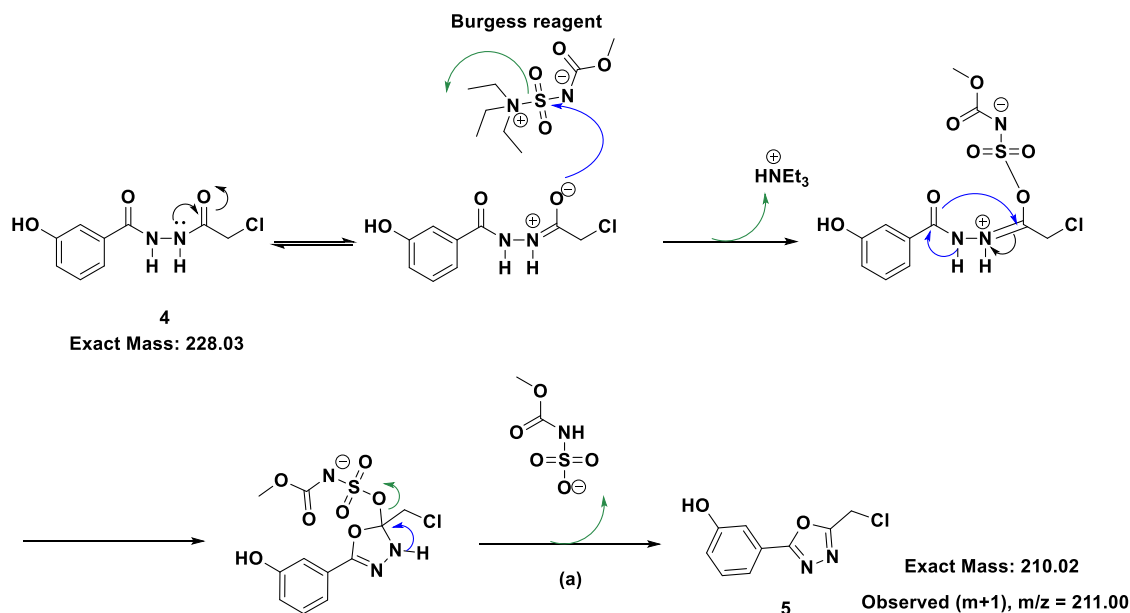
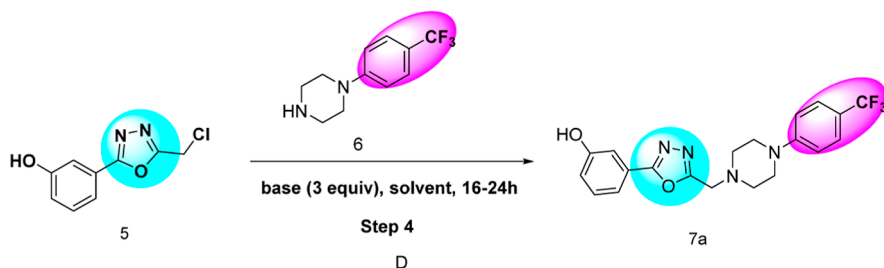


Figure 3. Plausible mechanism of oxadiazole ring formation.

nitrile oxides,²⁷ primary amides to nitriles,²⁸ and formamides to isonitriles,²⁹ have been achieved with Burgess reagent. We found that the reaction proceeded well using Burgess reagent (1.5 equiv) in dioxane at 100 °C. However, after 16 h, the reaction did not go to completion, as full conversion of the

starting material was not observed in TLC and LCMS. Only 41% conversion with 7% of Intermediate 4 was detected by LCMS. Complete conversion was achieved after 24 h, resulting in a 76% yield, as confirmed by TLC and LCMS (Figure 3). During the investigation of temperature and time, we

Table 2. Optimization of the Reaction Conditions for the Synthesis of 3-(5-((4-(4-(Trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenol (7a)^a



entry	base	solvent	temp	time (h)	% yield (%)
1	DIPEA	THF	rt	16	37
2	TEA	THF	rt	16	35
3	K ₂ CO ₃	THF	rt	24	29
4	DIPEA	THF	50 °C	16	48
5	TEA	THF	50 °C	16	46
6	K ₂ CO ₃	THF	50 °C	16	55
7	DIPEA	ACN	rt	16	36
8	TEA	ACN	rt	16	32
9	K ₂ CO ₃	ACN	rt	24	56
10	pyridine	THF	rt	16	81
11	pyridine	ACN	rt	16	77

^aReaction conditions: 3-(5-(chloromethyl)-1,2,4-oxadiazol-3-yl)phenol 5 (1 equiv), piperazine aryl derivatives 4 (1.1 equiv), base (3 equiv), solvent, 16–24 h.

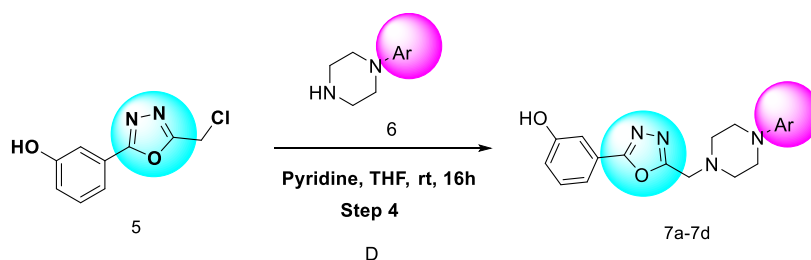
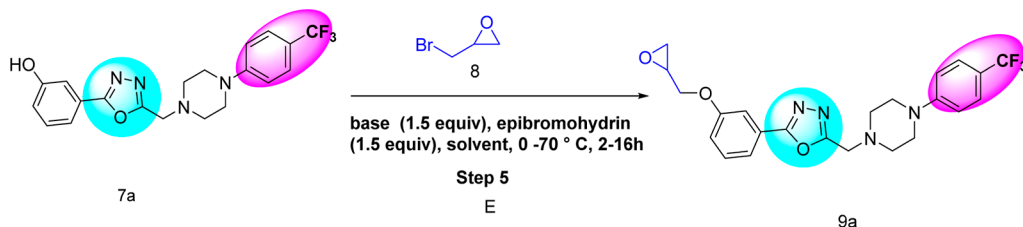


Figure 4. 3-(5-((4-Arylpiperazin-1-yl)methyl)-1,2,4-oxadiazol-3-yl)phenol derivatives (7a–7d).

Table 3. Optimization of the Reaction Conditions for the Synthesis of 2-(3-(Oxiran-2-ylmethoxy)phenyl)-5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazole 9a^a



entry	base	solvent	temp	time (h)	% yield
1	NaH	DMSO	0 °C–rt	16	13%
2	NaH	THF	0 °C–rt	2	87%
3	DIPEA	DCE	50 °C	16	
4	K ₂ CO ₃	ACN	70 °C	16	traces
5	K ₂ CO ₃	DMF	70 °C	16	traces
6	Cs ₂ CO ₃	ACN	70 °C	16	traces
7	Cs ₂ CO ₃	DMF	70 °C	36	14%

^aReaction conditions: 3-(5-((4-arylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenol derivatives (1 equiv), base (1.5 equiv), epibromohydrin (1.5 equiv), solvent, 0–70 °C, 2–16 h.

discovered that higher temperatures or longer reaction times led to lower yields. However, the yield improved when the solvent system was changed from THF to dioxane using

Burgess reagent (1.5 equiv). After completion the reaction mass was concentrated under reduced pressure. The obtained crude product was then purified by column chromatography,

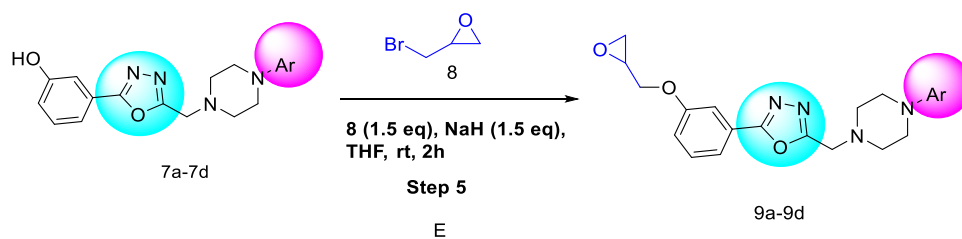
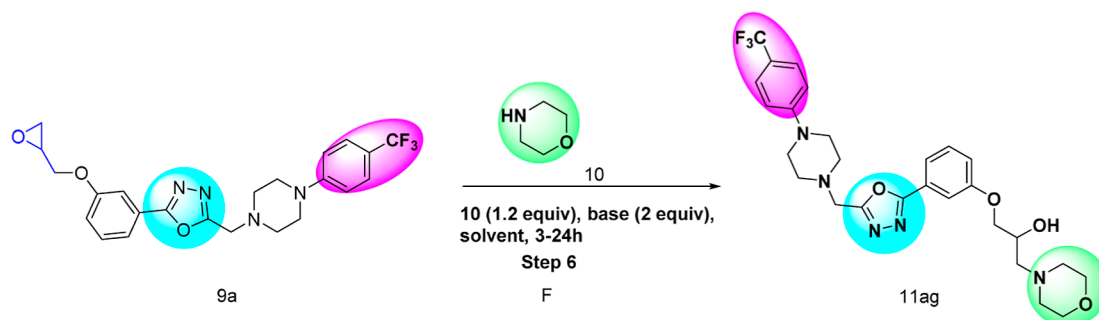


Figure 5. 5-((4-Arylpiperazin-1-yl)methyl)-3-(3-(oxiran-2-ylmethoxy)phenyl)-1,3,4-oxadiazole derivatives (9a–9d).

Table 4. Optimization of the Reaction Conditions for the Synthesis of 1-Morpholino-3-(3-(5-((4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,2,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11ag)^a



entry	base	solvent	temp (°C)	time (h)	% yield (%)
1		EtOH	90	24	36
2	TEA	THF	90	24	38
3	TEA	ACN	90	24	42
4	K ₂ CO ₃	ACN	90	3	94
5	pyridine	THF	90	24	42
6	DIPEA	DCE	90	24	44
7	DIPEA	IPA	90	24	56

^aReaction conditions: 5-((4-arylpiperazin-1-yl)methyl)-3-(3-(oxiran-2-ylmethoxy)phenyl)-1,2,4-oxadiazole (7a–7d) (1 equiv), 2° amine (1.2 equiv), base (2 equiv), solvent, 3–24 h.

eluting with 0–40% EtOAc in heptane. The desired fractions were concentrated and used for alkylation. Compound (5) was found to be unstable over time; even when stored for 24 h under freezing conditions (0–5 °C), the material deteriorated (Figures 2 and 3).

After the successful synthesis of (5), it was subjected to alkylation with piperazine aryl derivatives to identify the most effective reaction method for the synthesis of 3-(5-((4-arylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-2-yl)phenol derivatives (7a–7d). The reaction proceeded well and yielded good results when using 3 equiv of pyridine and 1.1 equiv of piperazine aryl derivatives in THF at room temperature (rt).

Numerous conditions were screened for the alkylation reaction of (7a), and it was determined that pyridine in THF provided the alkylated product in high yield (Table 2). During the investigation of temperature and time, we found that higher temperatures and longer reaction times resulted in lower yields under all conditions tested. The yield improved when the solvent system was changed from ACN to THF using pyridine as a base and maintaining the reaction at room temperature. The progress of the reaction was monitored via TLC and LCMS, and nearly complete conversion was observed after 16 h. The reaction mass was then concentrated under reduced pressure. The crude product was purified by column chromatography, eluting with 0–40% EtOAc in heptane (Figure 4). All the alkylated derivatives were synthesized using the same optimized conditions.

After the successful synthesis of 3-(5-((4-arylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenol derivatives (7a–7d), these compounds were reacted with epibromohydrin, resulting in Williamson ether synthesis and the formation of alkyl aryl ethers (9a–9d). Different optimization conditions were tested to identify the most feasible method for synthesizing (9a) (Table 3). It was found that the reaction proceeded well and gave a good yield when using NaH (1.5 equiv) and epibromohydrin (1.5 equiv) in THF at room temperature. Numerous conditions were screened for this O-alkylation reaction, and it was determined that NaH (1.5 equiv) in THF provided the alkyl aryl ether product in high yield.

During the investigation of the base and solvents, it was observed that reactions with other bases, such as K₂CO₃, DIPEA, and Cs₂CO₃, resulted in only traces of product under all conditions tested. The yield was found to increase when the solvent system was changed from ACN to THF using NaH as a base. The progress of the reaction was monitored via TLC and LCMS, and complete conversion was observed within 2 h. The reaction mass was then quenched with an aqueous solution of NH₄Cl and extracted with EtOAc. The organic layer was concentrated under reduced pressure, and the crude product was used directly for the next step of epoxy ring opening. All the O-alkylated derivatives were synthesized using the same optimized conditions (Figure 5).

After the successful synthesis of 5-((4-arylpiperazin-1-yl)methyl)-3-(3-(oxiran-2-ylmethoxy)phenyl)-1,3,4-oxadiazole

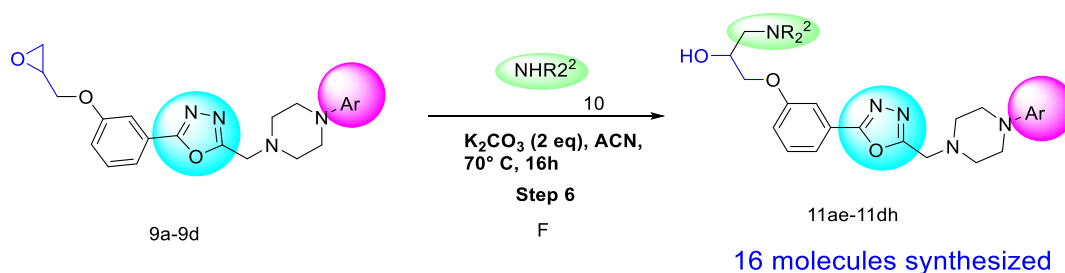
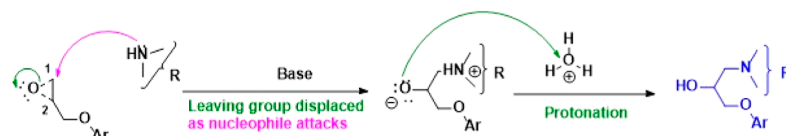


Figure 6. 5-((4-Arylpiperazin-1-yl)methyl)-3-(3-(oxiran-2-ylmethoxy)phenyl)-1,3,4-oxadiazole derivatives (**11ae–11dh**).

1) Possible ring opening, attack of nucleophile at the less substituted carbon C-1



2) Possible ring opening, attack of nucleophile at the more substituted carbon C-2

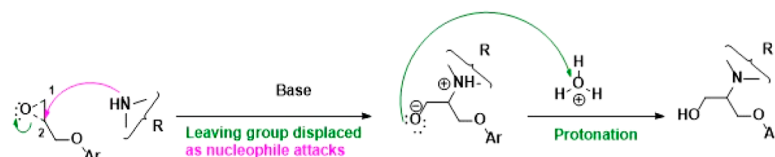


Figure 7. Plausible mechanism of epoxide ring opening.

derivatives, these compounds were treated with aliphatic amines to open the epoxy ring, resulting in the formation of 1-amino-3-(3-(5-((4-phenylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol.

Numerous conditions were screened for the oxirane ring opening reaction to identify the most feasible method for synthesizing (**11ae**) (Table 4). It was found that the reaction proceeded well and gave a good yield when using K_2CO_3 (2 equiv) and amine (1.1 equiv) in ACN at 90 °C. During the investigation of different bases and solvents, it was observed that reactions with other bases, such as TEA, DIPEA, and pyridine, gave only 36–56% yield under all tested conditions. The yield increased when the solvent system was changed from THF to ACN using K_2CO_3 as the base. The progress of the reaction was monitored via TLC and LCMS, and it was observed that the reaction showed complete conversion within 3 h. The reaction mass was then quenched with water and extracted with EtOAc. The organic layer was concentrated under reduced pressure, and the obtained crude product was purified by reverse-phase HPLC using the TFA method to afford the title compound. The desired fractions were concentrated and characterized by 1H NMR, ^{13}C NMR and HRMS (Figure 6).

The epoxide ring-opening reaction is well-documented using $BF_3 \cdot OEt_2$ or $AlCl_3$ as catalysts in alcohols for the synthesis of indoles from anilines and cyanoepoxides, as well as for the synthesis of anti- β -(*N*-arylamino)- α -hydroxynitriles by the regio- and diastereospecific ring-opening of 3-aryloxirane-2-carbonitriles with anilines. This process involves a tandem sequence, including the regiospecific ring-opening of cyanoepoxides with anilines, cyanide elimination, intramolecular aromatic electrophilic substitution, and water elimination.^{30–34} However, in our study, cyanoepoxides are not involved.

The **Int-9a–9d** compounds are epoxides with a more substituted carbon (C-2) and a less substituted carbon (C-1) attached to the oxygen. Typically, these epoxides yield a mixture of regio-isomers. When a strong nucleophile is used, it attacks the more accessible, less substituted carbon (C-1) from the opposite side, reacting faster, similar to the SN_2 mechanism. In contrast, unsymmetrical epoxides react at the more substituted carbon (C-2) when a weak nucleophile is used.³⁵

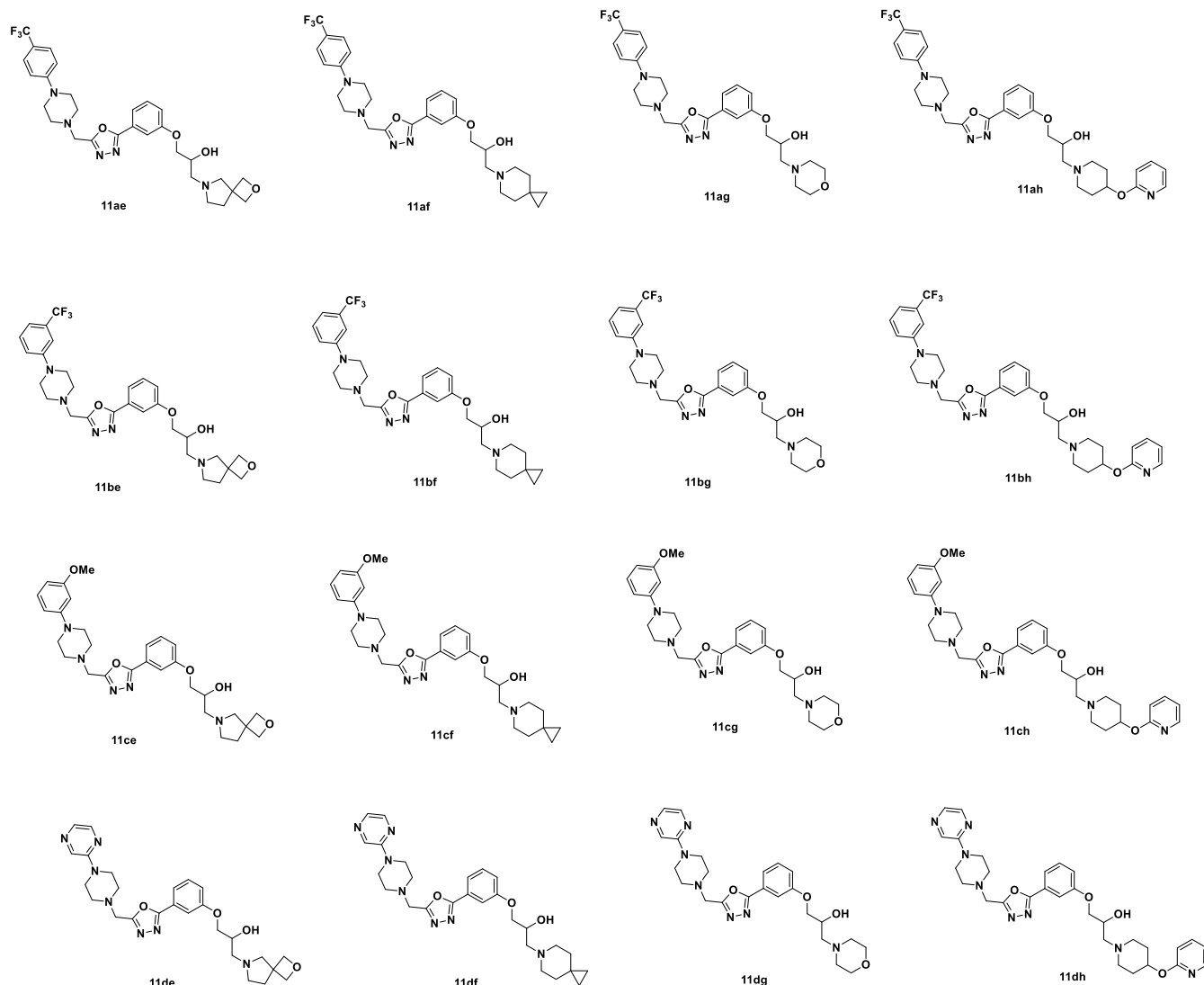
In our case, these epoxides, with a more substituted carbon (C-2) and a less substituted carbon (C-1), exclusively undergo attack at the C-1 position rather than at C-2, indicating that the steric effect dominates in this reaction (Figure 7).

To explore the generality of the method, different secondary amines were treated with **Int-9a–9d**, and we found that the yields were generally good, ranging from 65 to 94%. The results for the synthesis of 1-amino-3-(3-(5-((4-phenylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol are summarized in Table 5.

3. MATERIAL AND METHODS

3.1. Experimental Section. **3.1.1. Chemicals.** Chemical Co., Fluka, Spectrochem, Fisher Chemicals, BLD, Enamines, while solvents were procured from Finar, and Qualigens. Air and moisture-sensitive reactions were carried out under an inert atmosphere of argon in oven-dried glassware. Analytical thin-layer chromatography was performed on precoated TLC plates (layer 0.20 mm silica gel 60 with fluorescent indicator UV254, from Merck). Flash column chromatography was performed using prepacked silica gel cartridges (230–400 mesh, 40–63 μm , from Well Flash) using Biotage selekt or Combiflash Retrieve. 1H NMR was recorded on a Bruker Ascend 400 spectrometer (1H at 400 MHz). The correspond-

Table 5. Synthesis of 1-Amino-3-(3-(5-((4-phenylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol



ing solvents were DMSO and CDCl₃. All chemical shifts are given as δ values (parts per million) with reference to tetramethylsilane (TMS) as internal standard. LC–MS analyses were performed with either a Shimadzu–LC 40 DS connected to an SP 40 D Shimadzu photodiode array detector, or a Waters Acquity (QDa).

3.2. Gram Scale Synthesis of 2. To a solution of methyl 3-hydroxybenzoate (7 g, 46.05 mmol) in IPA (105 mL), 2-hydrazine hydrate (11.5 g, 230.25 mmol) was added slowly at room temperature. The reaction mixture was stirred for 16 h at 90 °C. The progress of the reaction was monitored by TLC and LCMS. Afterward, the reaction mixture was concentrated under reduced pressure to obtain the crude solid. The crude product was then stirred in 10% EtOAc in heptane for 20 min at room temperature. The white solid that precipitated out was filtered and dried under reduced pressure to afford the title compound (6.9 g, 97%) as a white solid. ¹H NMR (400 MHz, DMSO-*d*₆): 9.65 (s, 1H), 7.24–7.19 (m, 3H), 6.90–6.85 (m, 1H), 4.41–3.88 (m, 2H); ms: *m/z* 153.3 (*M* + 1).

3.3. Gram Scale Synthesis of 4. To a solution of (2) (5 g, 23.86 mmol) in THF (75 mL), chloroacetyl chloride (9.4 g, 36.15 mmol) was added slowly at 0 °C. The reaction mixture was stirred for 16 h at room temperature. The progress of the

reaction was monitored by TLC and LCMS. The white solid that precipitated out during the reaction was filtered, washed with *n*-heptane, and dried under reduced pressure to afford the title compound (7.2 g, 95%) as a white solid. ¹H NMR (400 MHz, DMSO-*d*₆): 10.39 (s, 1H), 10.32 (s, 1H), 9.75 (br s, 1H), 7.29–7.25 (m, 3H), 6.99–6.94 (m, 1H), 4.19 (s, 2H); ms: *m/z* 229.00 (*M* + 1).

3.4. Synthesis of 5. To a solution of (4) (2 g, 8.76 mmol) in dioxane (40 mL), Burgess reagent (3.14 g, 13.16 mmol) was added slowly at room temperature. The reaction mixture was stirred for 16 h at 100 °C. The progress of the reaction was monitored by TLC and LCMS. The reaction mixture was then concentrated under reduced pressure to obtain a crude solid. The crude product was purified by flash chromatography, eluting with 25–40% EtOAc in heptane, yielding 3-(5-(chloromethyl)-1,3,4-oxadiazol-2-yl)phenol as an off-white solid. The compound was found to be unstable and was immediately used in the next step (1.41 g, 76%). ¹H NMR (400 MHz, DMSO-*d*₆): 10.05 (br s, 1H), 7.45–7.38 (m, 3H), 7.05–7.02 (m, 1H), 5.13 (s, 2H); ms: *m/z* 211.05 (*M* + 1).

3.5. Gram Scale Synthesis of 7a–7d. **3.5.1. Synthesis of (7a).** To a solution of 3-(5-(chloromethyl)-1,3,4-oxadiazol-2-yl)phenol (1 g, 4.76 mmol) in THF pyridine (1.13 g, 14.7

mmol) and 1-(4-(trifluoromethyl)phenyl)piperazine (5.22 mmol) were added at rt. The reaction mixture was stirred for 16 h at rt. Progress of the reaction was monitored by TLC and LCMS. After completion, the reaction was diluted with water and extracted with EtOAc. The combined organics were dried over Na_2SO_4 , filtered and concentrated in vacuo. Obtained crude was purified by silica gel chromatography with 6:4 hexanes/EtOAc as the eluent gave the 3-(5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-2-yl)phenol (**7a**) as an off white solid (0.78 g, 81% yield); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 9.98 (br s, 1H), 7.51–7.45 (m, 2H), 7.44–7.38 (m, 3H), 7.07–7.00 (m, 3H), 3.97 (s, 2H), 3.33–3.28 (m, 4H), 2.69–2.67 (m, 4H); ms: m/z 405.17 ($M + 1$). All the homologous compounds were also prepared by a similar method.

3.6. Synthesis of 9a–9d. To a suspension of sodium hydride (55% in liquid paraffin, 2 equiv) in THF, 3-(5-((4-arylpiperazin-1-yl)methyl)-1,2,4-oxadiazol-3-yl)phenol (1 equiv) was added portion-wise at 0 °C. The reaction mixture was stirred at 0 °C for 10 min, followed by the dropwise addition of epibromohydrin (1.5 equiv) at the same temperature. The reaction mass was stirred at rt for 2 h. Progress of the reaction was monitored by TLC and LCMS. After completion, the reaction was quenched with aq solution of NH_4Cl and extracted with EtOAc. The combined organics were dried over Na_2SO_4 , filtered and concentrated in vacuo. Obtained crude was used as such for next step. All the other homologous compounds were also prepared by a similar method.

3.7. Synthesis of 11ae–11dh. To a solution of 5-((4-arylpiperazin-1-yl)methyl)-3-(3-(oxiran-2-ylmethoxy)phenyl)-1,3,4-oxadiazole (1 equiv) in ACN were added anhydrous K_2CO_3 (2 equiv) and amine (1.5 equiv) at rt. The reaction mixture was stirred for 3–8 h at 90 °C. Progress of the reaction was monitored by TLC and LCMS. After completion, the reaction was diluted with water and extracted with EtOAc. The combined organics were dried over Na_2SO_4 , filtered and concentrated in vacuo. Obtained crude was purified by reverse phase HPLC in TFA method to afford the title compound. All the homologous compounds were also prepared by a similar method.

3.8. Characterization of Products. **3.8.1. 3-(5-((4-(3-(Trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenol (**7b**).** Off white solid (0.77 g, 80% yield; eluent, 1/25 ethyl acetate/*n*-heptane); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 9.98 (s, 1H), 7.45–7.38 (m, 3H), 7.21 (d, $J = 8.4$ Hz, 1H), 7.16 (br s, 1H), 7.06 (d, $J = 7.6$ Hz, 1H), 7.02–6.99 (m, 2H), 3.97 (s, 2H), 3.27–3.25 (m, 4H), 2.71–2.68 (m, 4H); ms: m/z 405.18 ($M + 1$).

3.8.2. 3-(5-((4-(3-Methoxyphenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenol (7c**).** Off white solid (0.68 g, 78% yield; eluent, 1/25 ethyl acetate/*n*-heptane); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 9.98 (br s, 1H), 7.47–7.39 (m, 3H), 7.04–7.00 (m, 1H), 6.97–6.88 (m, 4H), 3.95 (s, 2H), 3.76 (s, 3H), 2.99 (br s, 1H), 2.68–2.67 (m, 4H); ms: m/z 367.15 ($M + 1$).

3.8.3. 3-(5-((4-(Pyrazin-2-yl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenol (7d**).** Off white solid (0.55 g, 65% yield; eluent, 1/40 ethyl acetate/*n*-heptane); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 9.97 (br s, 1H), 8.31 (d, $J = 1.4$ Hz, 1H), 8.08–8.07 (m, 1H), 7.82 (d, $J = 2.6$ Hz, 1H), 7.44–7.38 (m, 3H), 7.03–6.99 (m, 1H), 3.96 (s, 2H), 3.61–3.58 (m, 4H), 2.67–2.63 (m, 4H); ms: m/z 339.17 ($M + 1$).

3.8.4. 2-(3-(Oxiran-2-ylmethoxy)phenyl)-5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazole (9a**).** Light brown solid (0.8 g, Crude); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.62–7.59 (m, 1H), 7.55–7.48 (m, 4H), 7.26–7.23 (m, 1H), 7.06 (d, $J = 8.7$, 2H), 4.47 (dd, $J = 11.3$, 2.4 Hz, 1H), 3.97–3.91 (m, 3H), 3.32–3.26 (m, 5H), 2.88–2.85 (m, 1H), 2.76–2.75 (m, 1H), 2.72–2.69 (m, 4H); ms: m/z 461.0 ($M + 1$).

3.8.5. 3-(3-(Oxiran-2-ylmethoxy)phenyl)-5-((4-(3-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazole (9b**).** Light brown solid (0.80 g, Crude); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.61–7.60 (m, 1H), 7.56–7.50 (m, 2H), 7.40 (t, $J = 8.0$ Hz, 1H), 7.26–7.21 (m, 2H), 7.17 (br s, 1H), 7.07 (d, $J = 7.6$, 1H), 4.47 (dd, $J = 11.3$, 2.4 Hz, 1H), 3.98–3.91 (m, 3H), 3.73–3.70 (m, 3H), 3.45–3.41 (m, 2H), 2.89–2.85 (m, 1H), 2.76–2.74 (m, 1H), 2.73–2.68 (m, 4H); ms: m/z 461.0 ($M + 1$).

3.8.6. 5-((4-(3-Methoxyphenyl)piperazin-1-yl)methyl)-3-(3-(oxiran-2-ylmethoxy)phenyl)-1,3,4-oxadiazole (9c**).** Light brown solid (0.7 g, Crude); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.62–7.60 (m, 1H), 7.56–7.52 (m, 2H), 7.25 (dd, $J = 8.2$, 2.2 Hz, 1H), 6.96–6.84 (m, 4H), 4.47 (dd, $J = 11.3$, 2.0 Hz, 1H), 3.96 (s, 2H), 3.76–3.79 (m, 4H), 3.38–3.33 (m, 1H), 2.98 (br s, 4H), 2.88–2.85 (m, 1H), 2.77–2.75 (m, 1H), 2.68 (br s, 4H); ms: m/z 423.0 ($M + 1$).

3.8.7. 3-(3-(Oxiran-2-ylmethoxy)phenyl)-5-((4-(pyrazin-2-yl)piperazin-1-yl)methyl)-1,3,4-oxadiazole (9d**).** Light brown solid (0.6 g, Crude); ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 8.32 (d, $J = 1.4$ Hz, 1H), 8.09–8.05 (m, 1H), 7.84 (d, $J = 2.6$ Hz, 1H), 7.44–7.32 (m, 3H), 6.99–6.94 (m, 1H), 4.48 (dd, $J = 11.3$, 2.0 Hz, 1H), 3.98 (s, 1H), 3.87 (s, 2H), 3.38–3.35 (m, 1H), 3.27–3.25 (m, 1H), 2.97 (br s, 4H), 2.90–2.86 (m, 1H), 2.69 (br s, 4H); ms: m/z 395.0 ($M + 1$).

3.8.8. 1-(2-Oxa-6-azaspiro[3.4]octan-6-yl)-3-(3-(5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11ae**).** Off white solid, 92 mg, 74% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.58–7.56 (m, 1H), 7.53–7.48 (m, 4H), 7.22–7.19 (m, 1H), 7.06 (d, $J = 8.8$ Hz, 2H), 4.46 (s, 4H), 4.08–4.06 (m, 1H), 3.97–3.92 (m, 4H), 3.32–3.29 (m, 4H), 2.84–2.76 (m, 2H), 2.69–2.67 (m, 4H), 2.62–2.58 (m, 1H), 2.45–2.40 (m, 1H), 2.00 (t, $J = 7.2$ Hz, 2H), 1.89 (s, 3H); ^{13}C { ^1H } (100 MHz, $\text{DMSO}-d_6$): 164.3, 163.6, 159.2, 153.1, 130.73, 126.1, 126.08, 124.4, 118.7, 118.5, 118.0, 117.7, 114.2, 112.0, 82.3, 71.2, 67.5, 65.1, 58.52, 53.9, 51.7, 50.9, 46.8, 44.4, 35.7, 21.2; HRMS [$M + \text{H}$] $^+$ calcd for $\text{C}_{29}\text{H}_{34}\text{F}_3\text{N}_5\text{O}_4$, 574.2647; found, 574.2620.

3.8.9. 1-(6-Azaspiro[2.5]octan-6-yl)-3-(3-(5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11af**).** Off white solid, 110 mg, as a TFA salt, 89% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.58–7.47 (m, 5H), 7.24–7.21 (m, 1H), 7.05 (d, $J = 8.8$ Hz, 2H), 4.15–4.12 (m, 1H), 4.05–3.99 (m, 4H), 3.69–3.61 (m, 1H), 3.35–3.33 (m, 4H), 2.75–2.72 (m, 4H), 2.54–2.51 (m, 2H), 2.45–2.37 (m, 3H), 1.77–1.73 (m, 1H), 1.37–1.33 (m, 4H), 0.27–0.24 (m, 4H); ^{13}C { ^1H } (100 MHz, $\text{DMSO}-d_6$): 164.3, 163.6, 159.2, 153.1, 130.8, 126.3, 126.1, 124.4, 118.8, 118.5, 118.0, 117.7, 114.2, 112.0, 71.4, 65.5, 61.3, 53.6, 51.7, 50.9, 46.9, 34.7, 17.4, 11.1; HRMS [$M + \text{H}$] $^+$ calcd for $\text{C}_{30}\text{H}_{36}\text{F}_3\text{N}_5\text{O}_3$, 572.2841; found, 572.2834.

3.8.10. 1-Morpholino-3-(3-(5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11ag). Off white solid, 125 mg, 94% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.59–7.49 (m, 5H), 7.24–7.21 (m, 1H), 7.06 (d, J = 8.8 Hz, 2H), 4.98 (br s, 1H), 4.11–4.08 (m, 1H), 4.02–3.43 (m, 4H), 3.58–3.55 (m, 4H), 3.32–3.30 (m, 4H), 2.70–2.67 (m, 4H), 2.46–2.38 (m, 6H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 65.4, 163.2, 159.2, 153.2, 130.4, 126.5, 126.5, 124.9, 119.8, 114.8, 112.5, 70.6, 67.0, 65.4, 61.0, 53.8, 52.6, 52.1, 48.0, 29.8; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{27}\text{H}_{32}\text{F}_3\text{N}_5\text{O}_4$, 548.2481; found, 548.2477.

3.8.11. 1-(4-(Pyridin-2-yloxy)piperidin-1-yl)-3-(3-(5-((4-(4-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11ah). Off white solid, 109 mg, 79% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 8.14–8.12 (m, 1H), 7.68–7.64 (m, 1H), 7.59–7.57 (m, 1H), 7.56–7.52 (m, 4H), 7.46–7.21 (m, 1H), 7.05 (d, J = 8.8 Hz, 2H), 6.94–6.91 (m, 1H), 6.76–6.74 (m, 1H), 5.00–4.98 (m, 2H), 4.11–4.09 (m, 1H), 3.98–3.95 (m, 4H), 3.34–3.29 (m, 4H), 2.89–2.71 (m, 2H), 2.69–2.64 (m, 4H), 2.50–2.37 (m, 2H), 2.33–2.37 (m, 2H), 1.97–1.90 (m, 2H), 1.65–1.63 (m, 2H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 164.3, 163.5, 159.2, 153.1, 146.7, 139.1, 130.7, 126.1, 124.4, 118.8, 118.5, 116.7, 114.2, 112.0, 111.2, 71.4, 70.3, 65.6, 60.7, 51.7, 51.5, 51.3, 50.9, 46.8, 30.8; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{33}\text{H}_{37}\text{F}_3\text{N}_6\text{O}_4$, 639.2901; found, 639.2889.

3.8.12. 1-(2-Oxa-6-azaspiro[3.4]octan-6-yl)-3-(3-(5-((4-(3-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11be). Off white solid, 83 mg, 67%; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 9.89 (br s, 1H), 7.62–7.59 (m, 1H), 7.58–7.55 (m, 2H), 7.43 (t, J = 8.0 Hz, 1H), 7.26–7.22 (m, 2H), 7.18 (br s, 1H), 7.08 (d, J = 7.7 Hz, 1H), 4.62–4.55 (m, 3H), 4.45 (t, J = 6.0 Hz, 2H), 4.24–4.17 (m, 4H), 4.06–7.03 (m, 3H), 3.99–3.93 (m, 2H), 3.61–3.60 (m, 2H), 3.43–3.38 (m, 1H), 3.18–3.16 (m, 1H), 2.87 (m, 4H), 2.45–2.41 (m, 1H), 2.33–2.28 (m, 1H), 2.07 (s, 1H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 164.3, 162.8, 158.6, 150.8, 130.9, 129.9, 129.7, 125.7, 124.4, 123.0, 119.2, 118.5, 118.8, 118.4, 114.8, 112.3, 111.0, 70.1, 64.8, 64.6, 63.8, 63.6, 63.3, 59.2, 56.9, 56.8, 56.6, 55.2, 53.6, 51.7, 50.5, 48.5, 48.3, 47.1, 29.0; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{29}\text{H}_{34}\text{F}_3\text{N}_5\text{O}_4$, 574.2631; found, 574.2629.

3.8.13. 1-(6-Azaspiro[2.5]octan-6-yl)-3-(3-(5-((4-(3-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11bf). Off white solid, 109 mg, as a TFA salt, 88% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.59–7.57 (m, 1H), 7.54–7.50 (m, 2H), 7.41 (t, J = 8.0 Hz, 1H), 7.24–7.21 (m, 2H), 7.16 (br s, 1H), 7.06 (d, J = 7.6 Hz, 1H), 4.90 (br s, 1H), 4.11 (d, J = 6.5 Hz, 1H), 4.00–3.96 (m, 4H), 3.27–3.25 (m, 4H), 2.71–2.68 (m, 4H), 2.47–2.40 (m, 4H), 1.78–1.76 (m, 1H), 1.31 (br s, 4H), 0.25–0.23 (m, 4H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 164.3, 163.6, 159.2, 151.1, 130.7, 129.9, 129.6, 125.7, 124.4, 123.0, 118.8, 118.5, 114.6, 112.0, 110.9, 110.9, 71.4, 66.5, 61.2, 53.6, 51.8, 50.9, 47.5, 47.5, 47.5, 34.7, 17.3, 11.1; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{30}\text{H}_{36}\text{F}_3\text{N}_5\text{O}_3$, 572.2840; found, 572.2842.

3.8.14. 1-Morpholino-3-(3-(5-((4-(3-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11bg). Off white solid, 124 mg, 93%

yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.59–7.57 (m, 1H), 7.54–7.50 (m, 2H), 7.41 (t, J = 8.0 Hz, 1H), 7.24–7.21 (m, 2H), 7.16 (br s, 1H), 7.07 (d, J = 7.6 Hz, 1H), 4.94 (br s, 1H), 4.11–4.09 (m, 1H), 3.99–3.96 (m, 4H), 3.57 (t, J = 4.4 Hz, 4H), 3.27–3.26 (m, 4H), 2.71–2.68 (m, 4H), 2.46–2.37 (m, 5H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 172.0, 164.3, 163.6, 159.1, 151.1, 130.7, 129.9, 129.6, 125.7, 124.4, 123.0, 118.8, 118.5, 114.7, 114.6, 114.6, 114.5, 111.9, 111.0, 110.9, 110.9, 110.8, 71.3, 66.2, 61.3, 54.0, 51.8, 50.9, 47.5, 21.0; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{27}\text{H}_{32}\text{F}_3\text{N}_5\text{O}_4$, 548.2435; found, 548.2439.

3.8.15. 1-(4-(Pyridin-2-yloxy)piperidin-1-yl)-3-(3-(5-((4-(3-(trifluoromethyl)phenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (11bh). Off white solid, 101 mg, 73% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 8.14–8.11 (m, 1H), 7.68–7.64 (m, 1H), 7.58–7.53 (m, 1H), 7.51–7.49 (m, 2H), 7.41 (t, J = 8.0 Hz, 1H), 7.23–7.19 (m, 2H), 7.15 (s, 1H), 7.07–7.04 (m, 1H), 6.94–6.90 (m, 1H), 6.76–6.74 (m, 1H), 4.98–4.96 (m, 2H), 4.1–4.09 (m, 1H), 3.97 (br s, 4H), 3.26–3.23 (m, 4H), 2.80–2.70 (m, 2H), 2.69–2.67 (m, 4H), 2.42–2.40 (m, 1H), 2.34–2.27 (m, 2H), 1.97–1.93 (m, 2H), 1.65–1.62 (m, 2H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 164.3, 163.6, 162.5, 159.2, 151.0, 146.7, 139.1, 130.7, 129.9, 129.9, 129.6, 125.7, 124.4, 118.7, 118.5, 116.7, 114.6, 112.0, 111.1, 110.9, 71.4, 71.3, 66.6, 60.7, 51.8, 51.5, 51.3, 50.9, 47.5, 30.8; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{33}\text{H}_{37}\text{F}_3\text{N}_6\text{O}_4$, 639.2911; found, 639.2894.

3.8.16. 1-(3-(5-((4-(3-Methoxyphenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-(2-oxa-6-azaspiro[3.4]octan-6-yl)propan-2-ol (11ce). Off white solid, 91 mg, 72% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.59–7.57 (m, 1H), 7.53–7.50 (m, 2H), 7.22–7.20 (m, 1H), 6.96–6.84 (m, 4H), 4.98–4.97 (m, 1H), 4.46 (s, 4H), 4.08–4.05 (m, 1H), 3.97–3.91 (m, 4H), 3.76 (s, 3H), 2.98 (br s, 4H), 2.84–2.76 (m, 2H), 2.68 (br s, 4H), 2.62–2.58 (m, 1H), 2.45–2.40 (m, 1H), 2.01 (t, J = 7.2 Hz, 2H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 164.7, 164.2, 159.7, 152.4, 141.5, 131.2, 124.9, 122.9, 121.2, 119.3, 119.0, 118.4, 112.5, 112.3, 82.8, 71.7, 68.0, 65.5, 59.0, 55.7, 54.4, 52.8, 51.8, 51.6, 50.4, 44.9, 36.2; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{29}\text{H}_{37}\text{N}_5\text{O}_5$, 536.2866; found, 536.2869.

3.8.17. 1-(3-(5-((4-(3-Methoxyphenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-(6-azaspiro[2.5]octan-6-yl)propan-2-ol (11cf). Off white solid, 112 mg, as a TFA salt, 89% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.60–7.58 (m, 1H), 7.54–7.50 (m, 2H), 7.24–7.21 (m, 1H), 6.97–6.84 (m, 4H), 4.12–4.09 (m, 1H), 4.0–3.96 (m, 4H), 3.76 (s, 3H), 2.99 (br s, 4H), 2.68 (br s, 4H), 2.51–2.37 (m, 5H), 1.32 (m, 4H), 0.43 (br s, 1H), 0.25–0.21 (m, 4H); ^{13}C $\{^1\text{H}\}$ (100 MHz, $\text{DMSO}-d_6$): 164.3, 163.7, 159.2, 156.2, 151.9, 141.0, 130.8, 124.5, 122.4, 120.7, 118.8, 118.5, 118.0, 112.0, 111.8, 71.5, 66.5, 61.0, 55.2, 53.6, 52.3, 51.13, 49.9, 34.7, 17.4, 11.1; HRMS $[\text{M} + \text{H}]^+$ calcd for $\text{C}_{30}\text{H}_{39}\text{N}_5\text{O}_4$, 534.3069; found, 534.3067.

3.8.18. 1-(3-(5-((4-(3-Methoxyphenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-morpholinopropan-2-ol (11cg). Off white solid, 112 mg, 93% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, $\text{DMSO}-d_6$): δ 7.60–7.58 (m, 1H), 7.54–7.50 (m, 2H), 7.24–7.21 (m, 1H), 6.97–6.84 (m, 4H), 4.95

(br s, 1H), 4.13–4.08 (m, 1H), 4.03–3.94 (m, 4H), 3.76 (s, 3H), 3.57 (t, $J = 4.6$ Hz, 4H), 2.99 (br s, 4H), 2.68 (br s, 4H), 2.51–2.36 (m, 6H); ^{13}C { ^1H } (100 MHz, DMSO- d_6): 164.3, 163.7, 159.2, 151.9, 141.0, 130.8, 124.5, 122.5, 120.7, 118.8, 118.6, 118.0, 119.9, 111.8, 71.3, 66.2, 61.3, 55.2, 54.0, 52.3, 51.1, 49.9; HRMS [$\text{M} + \text{H}$] $^+$ calcd for $\text{C}_{27}\text{H}_{35}\text{N}_5\text{O}_5$, 510.2706; found, 510.2705.

3.8.19. 1-(3-(5-((4-(3-Methoxyphenyl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-(4-(pyridin-2-yloxy)piperidin-1-yl)propan-2-ol (**11ch**). Off white solid, 91 mg, 76% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, DMSO- d_6): δ 8.14–8.12 (m, 1H), 7.67–7.664 (m, 1H), 7.59–7.57 (m, 1H), 7.52–7.50 (m, 2H), 7.24–7.22 (m, 1H), 6.94–6.84 (m, 5H), 6.77–6.73 (m, 1H), 5.00–4.94 (m, 1H), 4.12–4.08 (m, 1H), 4.00–3.96 (m, 4H), 3.75 (s, 3H), 2.98 (br s, 4H), 2.84–2.76 (m, 2H), 2.67 (br s, 4H), 2.43–2.40 (m, 1H), 2.38–2.28 (m, 2H), 1.97–1.95 (m, 2H), 1.89 (br s, 2H), 1.68–1.64 (m, 2H); ^{13}C { ^1H } (100 MHz, DMSO- d_6): 164.7, 164.1, 159.7, 152.4, 147.2, 141.5, 139.6, 131.2, 124.9, 122.9, 121.2, 119.3, 119.0, 118.4, 117.2, 112.4, 112.2, 111.6, 71.9, 70.8, 67.1, 61.2, 55.7, 52.8, 52.3, 52.0, 51.8, 50.4, 31.3; HRMS [$\text{M} + \text{H}$] $^+$ calcd for $\text{C}_{33}\text{H}_{40}\text{N}_6\text{O}_5$, 601.3116; found, 601.3114.

3.8.20. 1-(3-(5-((4-(Pyrazin-2-yl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-(2-oxa-6-azaspiro[3.4]octan-6-yl)propan-2-ol (**11de**). Off white solid, 87 mg, as a TFA salt, 68% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, DMSO- d_6): δ 9.77 (br s, 1H), 8.36 (s, 1H), 8.10 (br s, 1H), 7.86 (d, $J = 2.4$, 1H), 7.66–7.54 (m, 3H), 7.26–7.24 (m, 1H), 5.50 (br s, 1H), 4.68–4.56 (m, 2H), 4.46–4.43 (m, 1H), 4.37–4.08 (m, 3H), 4.08–4.07 (m, 2H), 3.99–3.95 (m, 1H), 3.66 (BS, 4H), 3.48–3.37 (m, 2H), 3.34–3.32 (m, 2H), 3.21–3.13 (m, 1H), 2.86 (br s, 4H), 2.46–2.43 (m, 1H), 2.33–2.28 (m, 1H); ^{13}C { ^1H } (100 MHz, DMSO- d_6): 165.0, 164.6, 161.8, 158.7, 158.2, 154.2, 141.4, 133.5, 132.8, 131.8, 131.5, 130.8, 129.8, 124.4, 120.0, 119.3, 119.3, 118.5, 113.3, 112.3, 70.2, 64.8, 64.6, 63.8, 63.6, 63.4, 59.1, 56.9, 56.9, 56.7, 55.1, 53.6, 51.3, 51.0, 50.1, 48.0, 48.6, 48.4, 41.2, 29.1, 29.0; HRMS [$\text{M} + \text{H}$] $^+$ calcd for $\text{C}_{26}\text{H}_{33}\text{N}_7\text{O}_4$, 508.2653; found, 508.2651.

3.8.21. 1-(3-(5-((4-(Pyrazin-2-yl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-(6-azaspiro[2.5]octan-6-yl)propan-2-ol (**11df**). Off white solid, 108 mg, 85% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, DMSO- d_6): δ 9.40 (br s, 1H), 8.36 (br s, 1H), 8.09 (br s, 1H), 7.87 (d, $J = 2.4$ Hz, 1H), 7.62–7.54 (m, 3H), 7.29–7.24 (m, 2H), 4.38–4.36 (m, 1H), 4.28 (br s, 2H), 4.06 (d, $J = 4.6$ Hz, 2H), 3.55–3.52 (m, 4H), 3.41–3.37 (m, 2H), 3.29–3.23 (m, 1H), 3.16–3.04 (m, 3H), 2.95 (s, 4H), 2.16 (t, $J = 11.7$ Hz, 1H), 2.05 (t, $J = 13.3$ Hz, 1H), 1.21–1.23 (m, 2H), 0.44–0.29 (m, 4H); ^{13}C { ^1H } (100 MHz, DMSO- d_6): 165.0, 164.5, 158.6, 158.3, 157.9, 154.3, 141.4, 132.8, 131.5, 130.9, 124.4, 119.3, 118.4, 118.1, 117.9, 112.3, 70.2, 63.4, 57.8, 53.6, 51.4, 50.9, 50.3, 43.2, 30.6, 30.4, 15.6, 11.2, 11.1; HRMS [$\text{M} + \text{H}$] $^+$ calcd for $\text{C}_{27}\text{H}_{35}\text{N}_7\text{O}_3$, 506.2864; found, 506.2863.

3.8.22. 1-Morpholino-3-(3-(5-((4-(pyrazin-2-yl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol (**11dg**). Off white solid, 116 mg, as a TFA salt, 91% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, DMSO- d_6): δ 9.82 (br s, 1H), 8.35 (d, $J = 1.4$ Hz 1H), 8.11–8.10 (m, 1H), 7.87 (d, $J = 2.6$ Hz, 1H), 7.63–7.55 (m, 3H), 7.28–7.25 (m, 1H), 4.38–4.31

(m, 1H), 4.22 (br s, 2H), 4.08 (d, $J = 4.8$ Hz, 1H), 4.02–3.94 (m, 2H), 3.83–3.65 (m, 7H), 3.50–3.47 (m, 2H), 3.41–3.30 (m, 2H), 3.27–3.13 (m, 2H), 2.89 (m, 4H); ^{13}C { ^1H } (100 MHz, DMSO- d_6): 165.0, 164.5, 158.6, 158.2, 158.2, 157.9, 157.6, 154.3, 153.7, 141.4, 133.6, 133.3, 132.8, 131.8, 131.5, 130.1, 129.8, 124.4, 120.0, 119.3, 118.4, 117.8, 114.9, 113.4, 112.3, 70.2, 70.1, 63.0, 63.0, 58.3, 58.3, 52.8, 52.7, 51.3, 51.0, 50.4, 50.2, 43.0, 41.2; HRMS [$\text{M} + \text{H}$] $^+$ calcd for $\text{C}_{24}\text{H}_{31}\text{N}_7\text{O}_4$, 482.2507; found, 482.2506.

3.8.23. 1-(3-(5-((4-(Pyrazin-2-yl)piperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)-3-(4-(pyridin-2-yloxy)piperidin-1-yl)propan-2-ol (**11dh**). Off white solid, 79 mg, as a TFA salt, 65% yield; isolated through reverse phase HPLC purification in TFA method. ^1H NMR (400 MHz, DMSO- d_6): δ 8.31 (s, 1H), 8.14–8.12 (m, 1H), 8.07–8.05 (m, 1H), 7.83–7.82 (m, 1H), 7.68–7.64 (m, 1H), 7.58–7.49 (m, 3H), 7.23–7.21 (m, 1H), 6.94–6.91 (m, 1H), 6.76–6.74 (m, 1H), 5.00–4.92 (m, 2H), 4.11–4.08 (m, 1H), 3.98–3.96 (m, 4H), 3.59–3.57 (m, 4H), 2.83–2.76 (m, 2H), 2.65–2.63 (m, 4H), 2.43–2.38 (m, 2H), 2.33–2.25 (m, 2H), 1.97–1.94 (m, 2H), 1.68–1.61 (m, 2H); ^{13}C { ^1H } (100 MHz, DMSO- d_6): 164.3, 163.5, 162.5, 159.2, 154.5, 146.8, 141.4, 139.1, 132.5, 131.3, 130.8, 124.4, 118.8, 118.5, 116.7, 112.0, 111.2, 71.4, 70.3, 66.6, 60.7, 51.6, 51.6, 50.4, 50.0, 43.8, 30.8; HRMS [$\text{M} + \text{H}$] $^+$ calcd for $\text{C}_{30}\text{H}_{36}\text{N}_8\text{O}_4$, 573.2945; found, 573.2944.

4. CONCLUSION

In this study, we successfully synthesized 3-(3-(5-((4-phenylpiperazin-1-yl)methyl)-1,3,4-oxadiazol-3-yl)phenoxy)propan-2-ol amines novels derivatives, featuring key structural components such as a piperazine ring, an oxadiazole moiety, and a phenoxypropanol backbone. The synthesis was carried out through a multistep process. The reaction conditions were optimized to achieve a high yield, with careful attention to purification and characterization, ensuring the structural integrity of the final compound. Characterization techniques such as ^1H NMR, HRMS, and mass spectrometry confirmed the successful synthesis of the target molecule.

In conclusion, the successful synthesis of these complex structures demonstrates the feasibility of constructing multifunctional molecules with potential pharmacological significance. Further research on their biological properties will help to unlock their full potential as a therapeutic candidate.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsomega.5c01290>.

Copies of the ^1H NMR and HRMS spectra of the compounds (PDF)

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Author Contributions

Deepak Kumar conceived and designed the experiments, performed the experiments, analyzed the data and wrote the manuscript. All authors have read and agreed to the published version of the manuscript.

Notes

The authors declare no competing financial interest.

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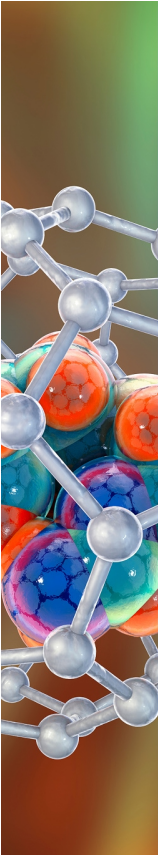
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